

Infosys TechCohere-DC Celebration Hackathon Challenges

ChennAI Crew – Smart Preordering and Billing for Food Court

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Smart Preordering and Billing for Food Court



Despite high footfall, current campus food-court systems rely on manual queues and disconnected billing. Pain points include:

- Long peak-hour queues are detrimental to productive time causing poor employee experience.
- Lack of real-time statistics for tracking consumption, wastage, and footfall.
- Limited analytics for demand forecasting, item velocity, and peak-hour staffing.
- Fragmented tracking of employee benefits and vendor sales which may cause reconciliation overhead.

Context, Business Relevance, Business Solution

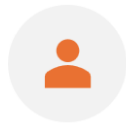
Context



Infosys campuses



High footfall during lunchtime



Multiple vendors



Heterogeneous menus

Business Relevance



Productivity: reduce time lost in queues; predictable pickups.



Revenue Optimization for vendors: lower wastage; align prep to demand.



Compliance & trust: prevent overcharging; auditable payments.



Operational efficiency: Centralized, auditable view of all vendors, sales, and consumption patterns; Data-driven space planning.

Business Solution

A unified digital food court platform with two personas (employee & vendor) that lets employees see live vendor availability, preorder, pay via UPI, and pick up with secure QR/OTP and accurate ETAs, backed by prepared-stock-aware billing to prevent overcharging. Azure OpenAI and ML personalize menu suggestions, generate product/nutrition info, interpret customizations, and forecast demand so vendors and facilities reduce wastage, improve efficiency, and enhance employee experience.

TECHNOLOGY STACK



Frontend: React.js &
React Native + Tailwind
CSS



Backend: Python +
FastAPI



Auth: Entra ID



Storage: PostgreSQL ,
Redis (cache) , S3 Bucket
(images & receipts)



ML Models: Prophet &
XGBoost



AI Model: GPT-4 (Azure
OpenAI)



Notifications: Firebase
Cloud messaging

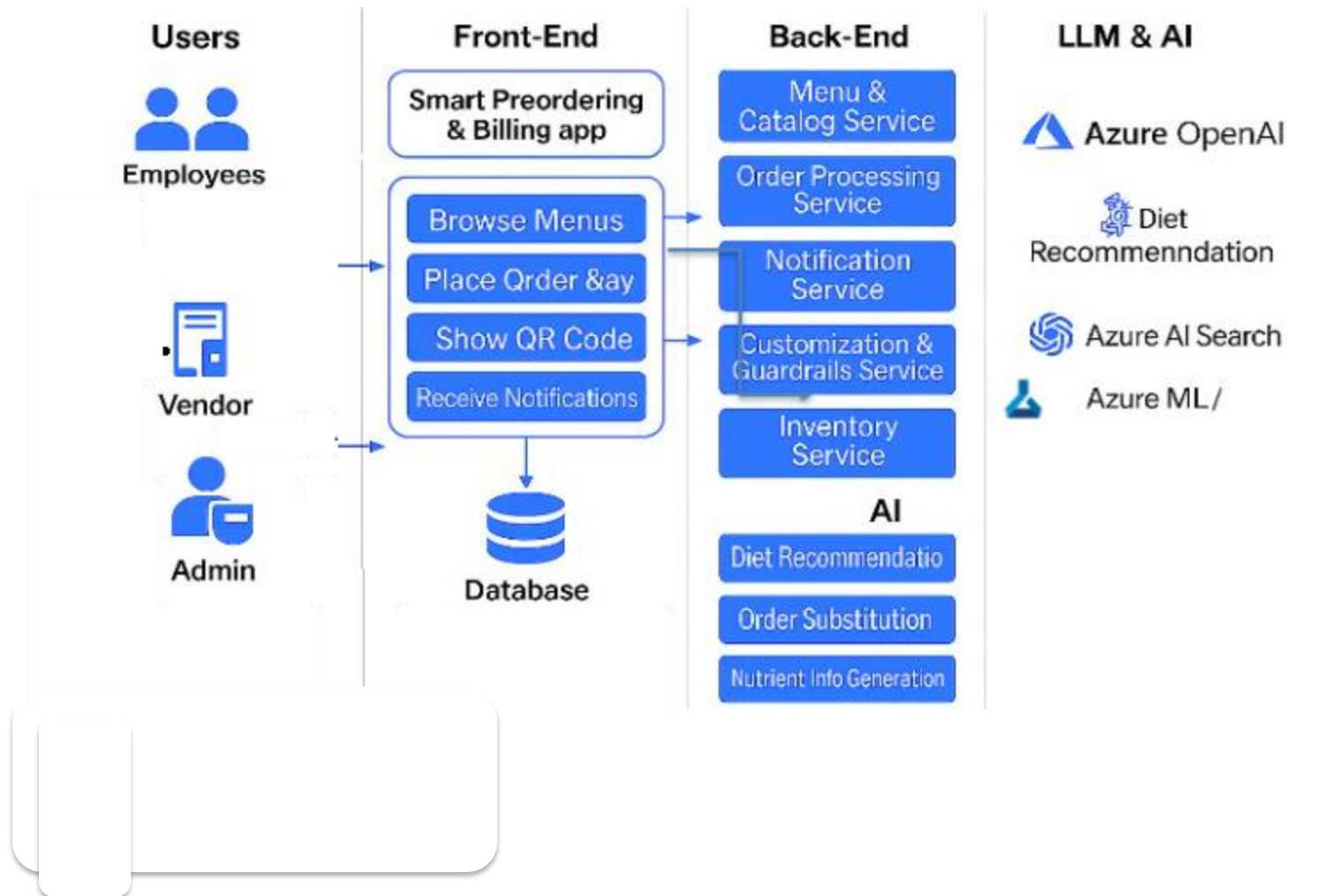


Hosting: Render



Payment: UPI gateway

Solution Architecture



KEY HIGHLIGHTS OF SOLUTION

Smart Preordering & Digital Billing: Employees browse real-time menus, place orders via UPI, and receive a digital bill with “QR + OTP” for secure pickup. Live ETA and Notifications feature available.

AI-powered Food Concierge (Azure OpenAI): Application collects diet preferences and current appetite information from customer via chat function in order to recommend food items. To protect customer’s privacy, they will have right to share the data to enhance the ML algorithm.

Smart Substitution suggester: If an item is missing or has run out of stock, Application suggests an appropriate substitute using GenAI.

AI-generated nutrient information: Nutrition information, ingredients used, description, etc. if not filled by the vendor will be generated by AI

Inventory planning: Using Machine learning models, the application forecasts the demand and presents the forecasted footfall and demand for the item every day (using historical sales, day, weather, events, festivals). It also generates number of ingredients that need to be ordered.

Customization request option: Employees will be able to add their preference or customization notes (for eg: no onion, more sambhar , etc) to their purchase and the vendor will be prompted to respond to the request. Sometimes, vendor can also request additional cost to be charged if required to fulfill the request.

Guardrails: Purchase thresholds will be set for each item and if ever customer would require to book more than the maximum limit, a permission would require to be taken.

Takeaway-ready Configuration: Items can be flagged as takeaway eligible, enabling planned office-to-home orders without hurting dine-in availability.

Ratings for menu items: Ratings will be collected and stored for the menu items in order to help customers make better decisions.

Analytics and governance layer: Dashboards for facilities and vendors: demand trends, wastage, fast-selling SKUs, peak hour heatmaps, threshold breaches, approvals; Insight loops from AI models continuously improve stocking, staffing, and menu decisions.

Future Enhancements



VisionAI to track wastage: Using Computer vision to capture images from garbage and track the wastage (for example: y grams of rice, x grams of dosa, etc.) and classify the waste for better monitoring.



Optimal Menu suggestor for vendors: Collecting diet information and preferences by users (only with consent) and using the information to suggest food choices and combinations to the vendors using Advanced AI.



Translator to local language: User's custom preferences could be translated/interpreted to the vendor in their local preferred language.



Voice-based ordering: Enablement of voice-based ordered in Indian languages using Azure Speech.



Anomaly detection to capture any possible threats to security.



Seat tracker: Using Computer vision identifying space available in the foodcourt.



Agentic AI to monitor hygiene and sanitation compliance keeping in track of cleanliness records and expiration dates of ingredients.

THANK YOU!